

Dr. Ishani I. Shah

Subject Matter Expert - Life Sciences | EdTech Content Developer | Researcher
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PROFESSIONAL SUMMARY

PhD Biotechnologist and EdTech Specialist with 12+ years of experience spanning academic research and the education industry. Expertise in developing and delivering 800+ life science e-learning modules across video, 3D simulation and AR/VR formats, aligned with CBSE, ICSE, NGSS, GSEB, and NEET curricula. Strong subject expertise in molecular biology, genetics, microbiology, and biochemistry. Proficient in leveraging AI tools for content development, scientific review, and quality assurance. Multilingual in English, Hindi, and Gujarati.

CORE COMPETENCIES

Curriculum Design · Instructional Design · Learning Experience & Assessment Design · Biology Content Specialist · Educational Technology · AR/VR & 3D Simulation · Generative AI Tools · AI-assisted Content Development & Review · Scientific Quality Assurance · STEM Education · Teacher Training · Product Development · Project Management · Database design and management · Molecular Biology · Biotechnology · Bioinformatics · Microbiology · Genetics · Biochemistry · Haematology · Plant Pathology · Scientific Research & Publications

WORK EXPERIENCE

Subject Matter Expert | Phibonacci Solutions Pvt. Ltd., Ahmedabad

2022 – Present

- Lead biology team to design and deliver 800+ research-based life science e-learning modules across video, 3D simulation, and AR/VR formats; oversee scripting, content creation, and translation.
- Use AI tools, including Claude, ChatGPT, and Gemini, to accelerate content drafting and iteration; review all AI-generated outputs for scientific accuracy before publication.
- Design and map curricula to CBSE, ICSE, NGSS, and GSEB standards; develop teacher training modules aligned to learner and stakeholder objectives.
- Manage full product lifecycle, including content planning, stakeholder engagement, multi-team coordination, and quality assurance.

Research Fellow | KSV University, Gandhinagar

2013 – 2022

- Investigated G6PD enzyme deficiency in 3,000+ subjects across Gujarat through field surveys, phlebotomy, and haematological and biochemical analysis.
- Discovered novel mutation (NCBI GenBank Accession: ON262108)
- Conducted sequence analysis using BLAST
- Performed variant effect prediction using SIFT, PolyPhen-2, PROVEAN, PANTHER, FATHMM, Meta-SNP, PhD-SNP, SNAP, SNP2, MA, and Mutation Assessor
- Modelled protein structure and mutation impact using FoldX via PremPS and BIOVIA Discovery Studio Visualizer
- Designed a dedicated mutation database
- Conducted molecular analysis, including DNA isolation and PCR, on clinical samples.
- Presented research at 10 national and international conferences in English, Hindi, and Gujarati.

Junior Research Fellow | S.P. University, Vallabh Vidyanagar

Dec 2015 – May 2016

- Molecular analysis of genes conferring TLCV tolerance in tomato varieties; GSBTM-funded project.

IBT Programme Fellow | GSBTM / Biocare Research India Pvt. Ltd.

Oct 2012 – Jan 2013

- Investigated antimicrobial activity of ortho-implant against *Staphylococcus aureus*; developed a clinical sample management database.
- Industrial microbiological testing to NABL standards; clinical biosafety and sterile technique.

Science Tutor | Alpha Science Academy Pvt. Ltd., Ahmedabad

2011 – 2012

- Designed study modules and conducted practical sessions and weekly examinations for Std. 11–12 PCB students.

EDUCATION

Doctor of Philosophy — Biotechnology | KSV University, Gandhinagar

2023

Master of Science — Bioinformatics | KSV University, Gandhinagar

2012

Bachelor of Science — Microbiology | Gujarat University, Ahmedabad

2010

PUBLICATIONS

1. Vaghela P, **Shah I**, Jarullah B. Rheumatoid Arthritis and Comorbid Conditions: is there A Missing Link?. Biosc.Biotech.Res. Comm. 2020;13(4).

2. **Shah II.**, Jarullah J., Jarullah B., Prevalence of G6PD Deficiency in India: A Systematic Review. Advances in Bioscience and Biotechnology. 2018, 9(09), 481-496. (Impact factor: 1.26)

3. **Shah I.**, Jarullah J., Jaouni SA., Jamal MS., Jarullah B., Higher prevalence of G6PD deficiency in tribal population against urban population: A signal to natural selection. Biomedical Research 2017, 28 (1) 134-139. (Impact factor: 3.06)

4. **Shah I.**, Jarullah B., Jamal MS., Jarullah J., Low versus high frequency of G6PD deficiency in urban against tribal population of Gujarat – a signal to natural selection. BMC Genomics, 2016, 17(6). (Impact factor: 3.7)

ACHIEVEMENTS

- Best Poster Presentation — National Symposium, BRD School of Biosciences, S.P. University (2016)
- Excellence in Research Award — Women's Development Cell, KSV University (2016)
- University Rank Holder (M.Sc.) — First Rank in Part I; Second Rank in Part II (2010-2012)
- 1st Prize, Inter-collegiate Science Quiz — R.G. Shah Science College (2010)

SKILLS

Languages: English (Professional) | Hindi (Fluent) | Gujarati (Native)

Tools: Microsoft Office | PHP & MySQL | Claude, ChatGPT, Gemini

Bioinformatics: BLAST | SIFT | PolyPhen-2 | PROVEAN | PANTHER | FATHMM | Meta-SNP | PhD-SNP | SNAP | SNP2 | MA | FoldX | PremPS | BIOVIA Discovery Studio Visualizer | Mutation Assessor

Wet Lab: Molecular biology techniques | Clinical microbiology | Phlebotomy | Haematological analysis